

TidyTuesday Week 37

Pranay Gundam

Sunday 13th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly edition of Two Random Data Series Walk Into A Bar, where we bravely regress things that have no business being in the same spreadsheet and then act shocked at the results. This week our tireless robot pulled seven pairs out of FRED's endless bin of numbers, and honestly, the lineup did not disappoint. We opened Sunday by asking whether Lithuania's "Discussion About Pandemics Index" – a series so obscure its popularity score is literally the number one, as in one lonely soul – has anything to do with the U.S. Treasury borrowing from the public. And, uh, kind of? The raw regression was significant, and remarkably it held up detrended too, which is the rare and unsettling case where I'm forced to stop dunking and just stare into the middle distance. Do Lithuanians chatter about pandemics whenever Uncle Sam floats more debt? No. Absolutely not. But the p-value doesn't know that, and that's the fun part.

Monday gave us coal-fired carbon dioxide emissions versus the price index for treating congenital anomalies, which is a sentence I never planned to write. Raw R-squared of about 0.61, and it actually survived detrending with an even punchier slope. Two things drifting together for decades usually means nothing, yet here the relationship refused to evaporate, so I'll grudgingly file it under "huh." Tuesday, meanwhile, was the classic trend mirage we live for: the market value of gross federal debt versus business equipment leases owned by finance companies. The raw regression insisted there was a link with a p-value with eleven zeros in it, but of course both series just spent decades marching upward like everything else denominated in dollars. Detrended, the relationship flipped sign and got even more "significant," which mostly tells you the whole thing is numerical noise wearing a nice suit.

Wednesday's pairing – construction quits versus the work-from-home rate for women – was the week's comedy relief. The raw regression was a spectacular flatline, an R-squared of 0.0004 and a p-value of 0.86, which is statistics for "these two have never met." Then, plot twist, detrending yanked out a genuinely significant negative relationship, proving that once you strip the trends away sometimes a ghost appears where there was nothing before. Thursday brought national wages and salaries against manufacturing employment in Ogden, Utah, a raw R-squared of 0.80 that shrank to a wobblier but still-present 0.40 detrended, so partial credit. And Friday delivered Midwest shelter costs versus the price of a whole chicken in the West, which posted a jaw-dropping raw R-squared of 0.94 – both rent and poultry go up over time, film at eleven – but still held on detrended, suggesting that when the cost of living climbs, so does the cost of dinner. Groundbreaking.

We closed Saturday on the week's most legitimately boring result, and I mean that as a com-

pliment: two employment cost indices, one for private production workers and one for state and local government, correlating at a raw R-squared of 0.99 and holding strong at 0.83 detrended. This is the one pairing all week where the relationship is actually plausible, because compensation costs really do move together across sectors – so naturally it's the least interesting to write about. As always, the mandatory disclaimer: these are randomly matched series, no theory, no causation, no rigor, just a machine flinging numbers at each other for our amusement. Spurious correlation is the house special here, not the exception, so please do not adjust your portfolio, your carbon policy, or your chicken budget based on anything above. See you next week for more statistically significant nonsense.

2 Date: 2026-09-07

Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MTSMFBP133FMS	R-squared:	0.199			
Model:	OLS	Adj. R-squared:	0.186			
Method:	Least Squares	F-statistic:	15.87			
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000176			
Time:	16:43:46	Log-Likelihood:	-893.46			
No. Observations:	66	AIC:	1791.			
Df Residuals:	64	BIC:	1795.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.886e+04	2.46e+04	1.985	0.051	-321.644	9.8e+04
value_fred_WPDILTU	1205.6312	302.613	3.984	0.000	601.092	1810.170
Omnibus:	51.230	Durbin-Watson:	2.093			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	275.412			
Skew:	2.139	Prob(JB):	1.57e-60			
Kurtosis:	12.047	Cond. No.	87.5			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

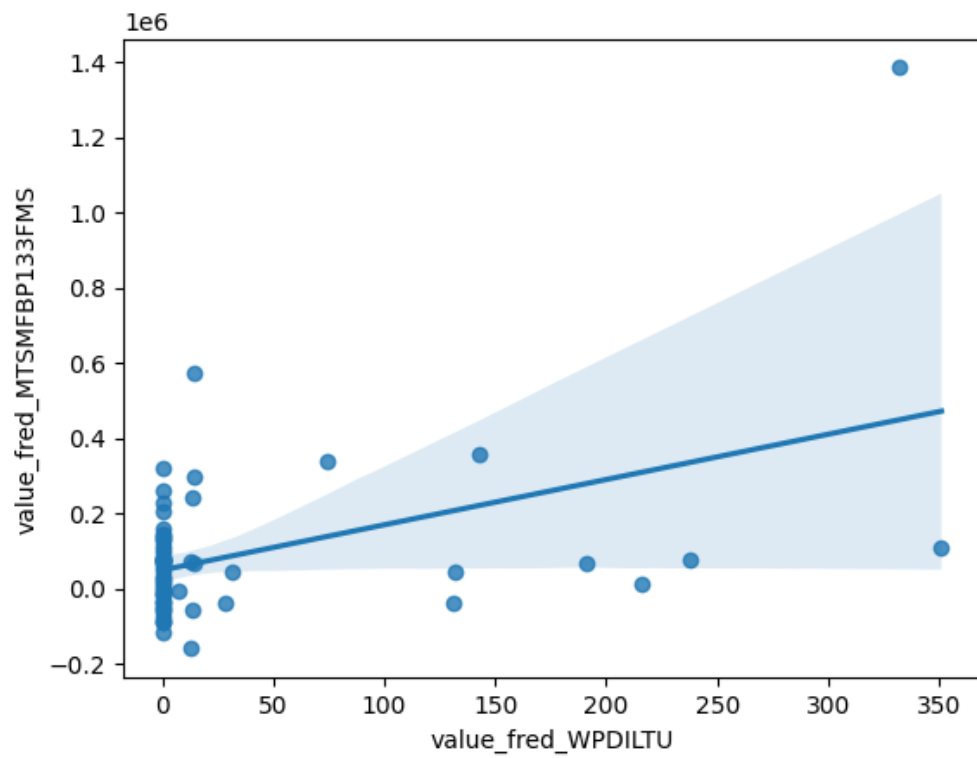


Figure 1: Raw Regression Plot for 2026-09-07

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MTSMFBP133FMS_detrended	R-squared:	0.174
Model:	OLS	Adj. R-squared:	0.161
Method:	Least Squares	F-statistic:	13.48
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000493
Time:	16:43:46	Log-Likelihood:	-893.06
No. Observations:	66	AIC:	1790.
Df Residuals:	64	BIC:	1794.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	5.051e-09	2.28e+04	2.22e-13	1.000	-4.55e+04	4.55e+04
value_fred_WPDILTU_detrended	1138.7087	310.105	3.672	0.000	519.204	1758.214

Omnibus:	52.012	Durbin-Watson:	2.129
Prob(Omnibus):	0.000	Jarque-Bera (JB):	302.597
Skew:	2.138	Prob(JB):	1.96e-66
Kurtosis:	12.579	Cond. No.	73.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

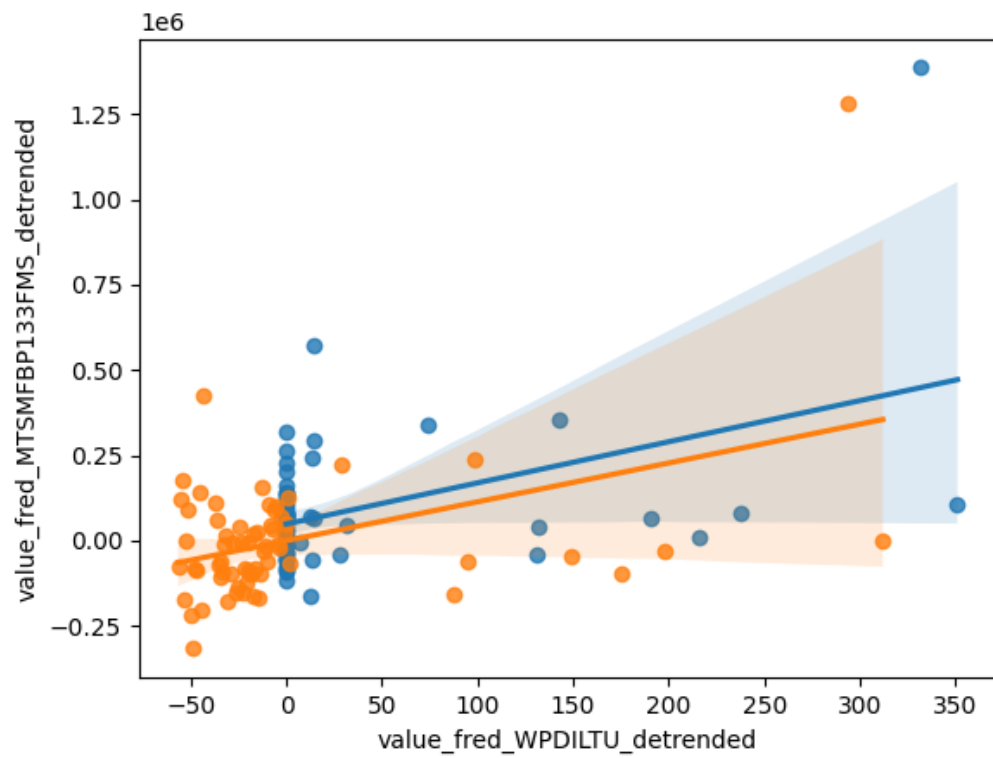


Figure 2: Detrended Regression Plot for 2026-09-07

3 Date: 2026-09-08

Series ID: EMISSCO2TOTVTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_COGANOPIMEPS	R-squared:	0.606
Model:	OLS	Adj. R-squared:	0.585
Method:	Least Squares	F-statistic:	29.22
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	3.24e-05
Time:	15:27:44	Log-Likelihood:	-111.42
No. Observations:	21	AIC:	226.8
Df Residuals:	19	BIC:	228.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-146.4574	51.777	-2.829	0.011	-254.827	-38.088
value_fred_EMISSCO2TOTVTTCOUSA	0.1533	0.028	5.406	0.000	0.094	0.213

Omnibus:	3.022	Durbin-Watson:	1.167
Prob(Omnibus):	0.221	Jarque-Bera (JB):	1.530
Skew:	0.625	Prob(JB):	0.465
Kurtosis:	3.429	Cond. No.	8.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

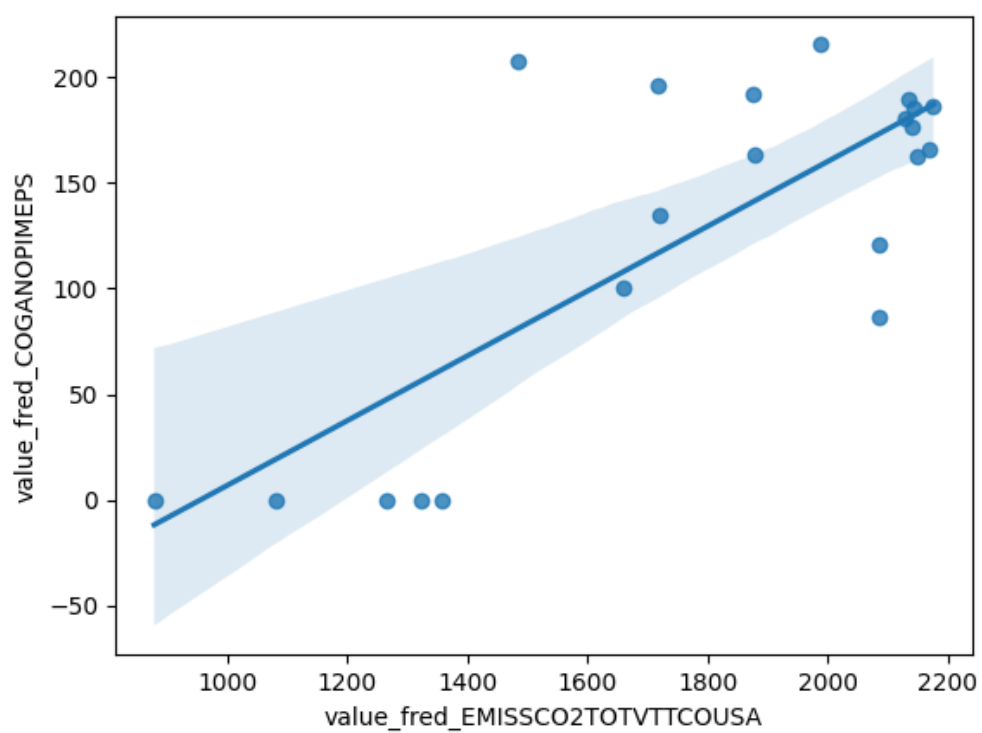


Figure 3: Raw Regression Plot for 2026-09-08

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_COGANOPIMEPS_detrended	R-squared:	0.559				
Model:	OLS	Adj. R-squared:	0.536				
Method:	Least Squares	F-statistic:	24.09				
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	9.76e-05				
Time:	15:27:44	Log-Likelihood:	-108.14				
No. Observations:	21	AIC:	220.3				
Df Residuals:	19	BIC:	222.4				
Df Model:	1						
Covariance Type:	nonrobust						
		coef	std err	t	P> t	[0.025	0.975
const		-6.734e-12	9.567	-7.04e-13	1.000	-20.023	20.023
value_fred_EMISSCO2TOTVTTCOUSA_detrended		0.3032	0.062	4.908	0.000	0.174	0.432
Omnibus:		4.692	Durbin-Watson:	1.541			
Prob(Omnibus):		0.096	Jarque-Bera (JB):	2.578			
Skew:		0.726	Prob(JB):	0.276			
Kurtosis:		3.915	Cond. No.	155.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

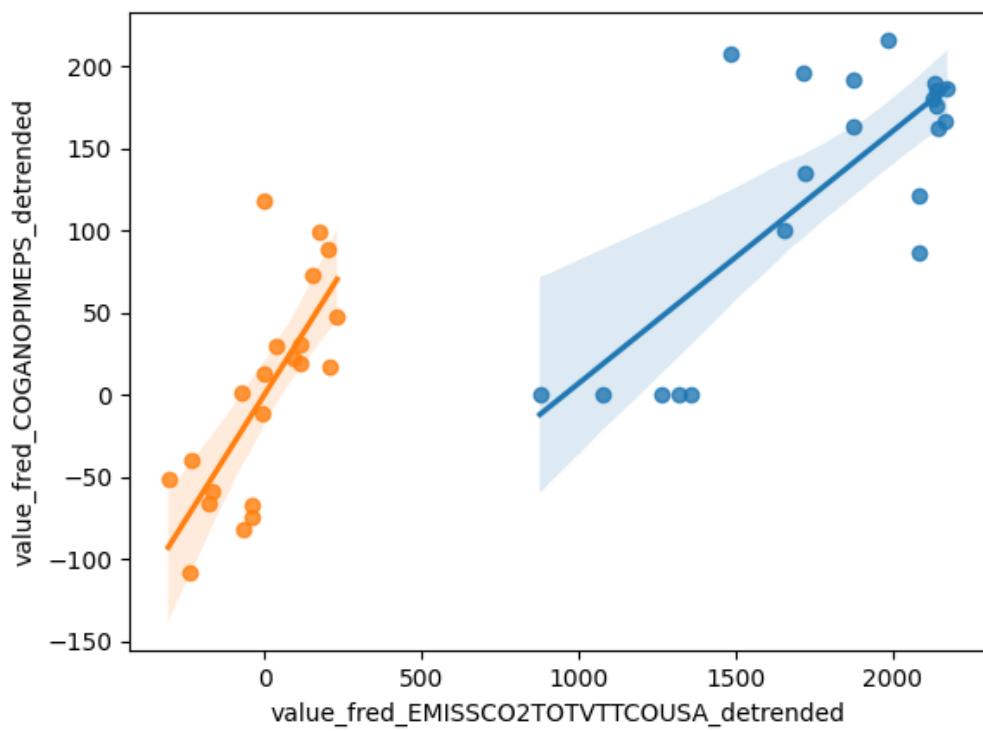


Figure 4: Detrended Regression Plot for 2026-09-08

4 Date: 2026-09-09

Series ID: MVGFD027MNFRBDAL

This series is titled Market Value of Gross Federal Debt and has a frequency of Monthly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1942-01-01 and the observation end date is 2026-07-01. The popularity of this series is 37.

Series ID: DTBTERNM

This series is titled Business Equipment Leases Owned and Securitized by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DTBTERNM	R-squared:	0.095
Model:	OLS	Adj. R-squared:	0.093
Method:	Least Squares	F-statistic:	45.33
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	5.34e-11
Time:	15:24:33	Log-Likelihood:	-5318.4
No. Observations:	433	AIC:	1.064e+04
Df Residuals:	431	BIC:	1.065e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.633e+05	4472.957	36.517	0.000	1.55e+05	1.72e+05
value_fred_MVGFD027MNFRBDAL	-1.7290	0.257	-6.732	0.000	-2.234	-1.224

Omnibus:	2749.157	Durbin-Watson:	0.021
Prob(Omnibus):	0.000	Jarque-Bera (JB):	48.079
Skew:	0.054	Prob(JB):	3.63e-11
Kurtosis:	1.371	Cond. No.	3.10e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.

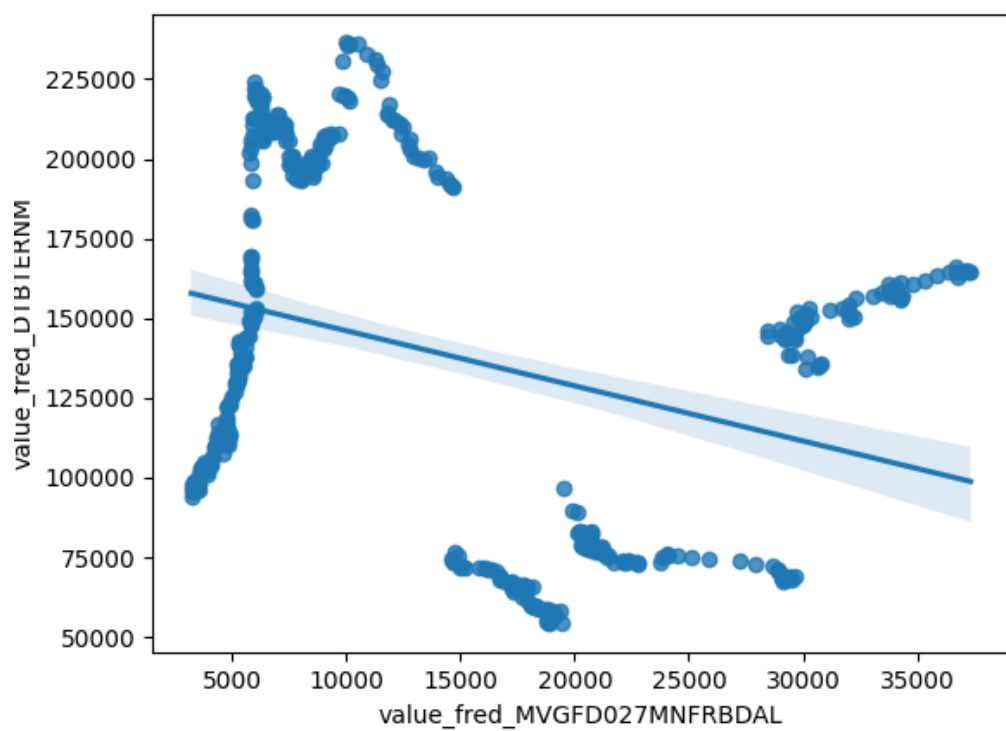


Figure 5: Raw Regression Plot for 2026-09-09

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DTBTERNM_detrended	R-squared:	0.116
Model:	OLS	Adj. R-squared:	0.114
Method:	Least Squares	F-statistic:	56.72
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	2.97e-13
Time:	15:24:33	Log-Likelihood:	-5302.8
No. Observations:	433	AIC:	1.061e+04
Df Residuals:	431	BIC:	1.062e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	9.044e-09	2427.545	3.73e-12	1.000	-4771.300	4771.300
value_fred_MVGFD027MNFRBDAL_detrended	-6.1487	0.816	-7.531	0.000	-7.753	-4.543

Omnibus:	698.313	Durbin-Watson:	0.023
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33.214
Skew:	-0.174	Prob(JB):	6.13e-08
Kurtosis:	1.689	Cond. No.	2.97e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.97e+03. This might indicate that there are strong multicollinearity or other numerical problems.

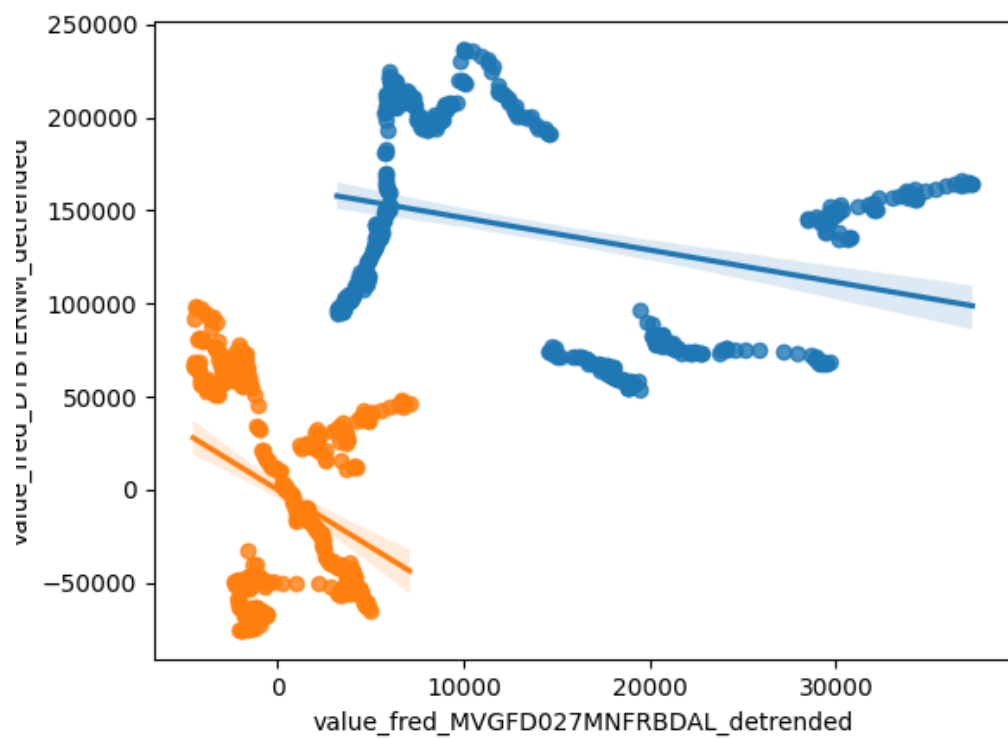


Figure 6: Detrended Regression Plot for 2026-09-09

5 Date: 2026-09-10

Series ID: JTS2300QUR

This series is titled Quits: Construction and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 13.

Series ID: WFHCOVIDFRACMATWOMEN

This series is titled Work from Home Rate: Women and has a frequency of Monthly. The units are Percent of Full Paid Working Days and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2020-05-01 and the observation end date is 2026-08-01. The popularity of this series is 6.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.013
Method:	Least Squares	F-statistic:	0.03323
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	0.856
Time:	15:17:47	Log-Likelihood:	-240.33
No. Observations:	74	AIC:	484.7
Df Residuals:	72	BIC:	489.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	32.4960	3.704	8.774	0.000	25.113	39.879
value_fred_JTS2300QUR	-0.3289	1.804	-0.182	0.856	-3.926	3.268

Omnibus:	71.083	Durbin-Watson:	0.194
Prob(Omnibus):	0.000	Jarque-Bera (JB):	449.811
Skew:	2.992	Prob(JB):	2.11e-98
Kurtosis:	13.492	Cond. No.	12.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

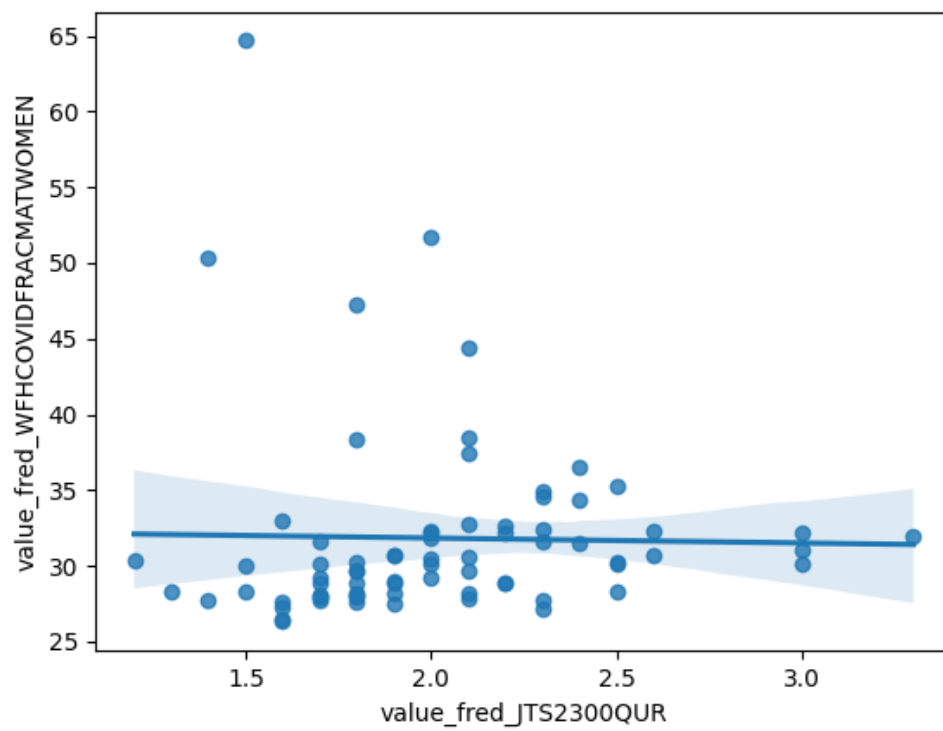


Figure 7: Raw Regression Plot for 2026-09-10

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN_detrended	R-squared:	0.300
Model:	OLS	Adj. R-squared:	0.290
Method:	Least Squares	F-statistic:	30.84
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	4.44e-07
Time:	15:17:47	Log-Likelihood:	-205.78
No. Observations:	74	AIC:	415.6
Df Residuals:	72	BIC:	420.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.578e-12	0.460	-1.65e-11	1.000	-0.917	0.917
value_fred_JTS2300QUR_detrended	-7.2666	1.309	-5.553	0.000	-9.875	-4.658

Omnibus:	45.009	Durbin-Watson:	0.681
Prob(Omnibus):	0.000	Jarque-Bera (JB):	171.162
Skew:	1.828	Prob(JB):	6.80e-38
Kurtosis:	9.492	Cond. No.	2.84

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

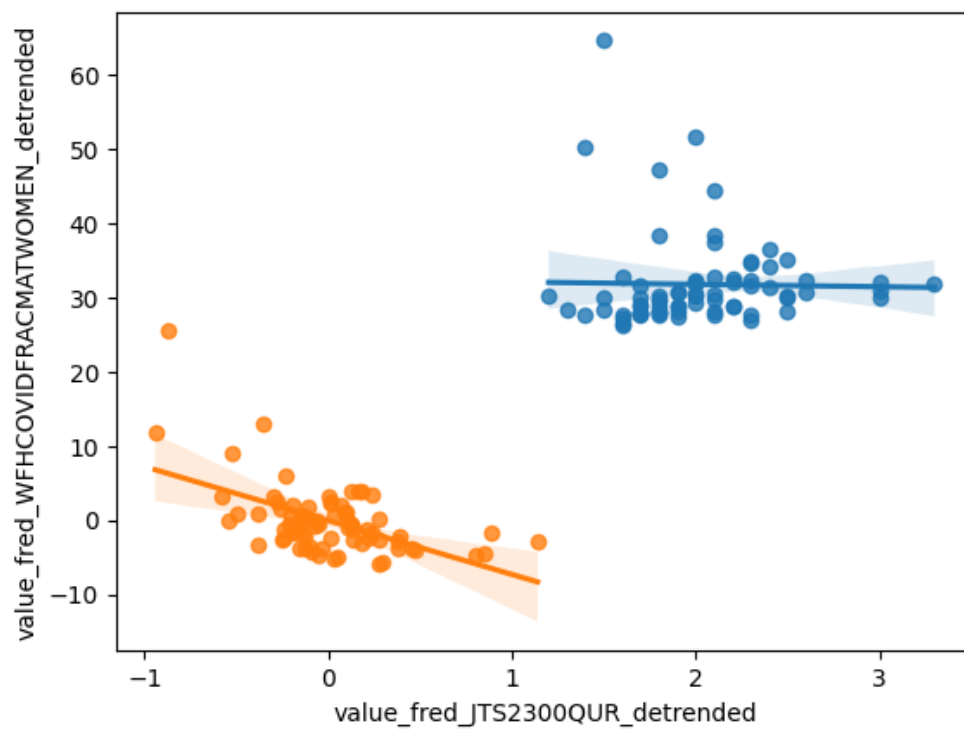


Figure 8: Detrended Regression Plot for 2026-09-10

6 Date: 2026-09-11

Series ID: A034RC1A027NBEA

This series is titled National income: Compensation of employees: Wages and salaries and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2025-01-01. The popularity of this series is 23.

Series ID: OGDE249MFGN

This series is titled All Employees: Manufacturing in Ogden, UT (MSA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OGDE249MFGN	R-squared:	0.802
Model:	OLS	Adj. R-squared:	0.797
Method:	Least Squares	F-statistic:	138.0
Date:	Fri, 11 Sep 2026	Prob (F-statistic):	1.62e-13
Time:	15:19:38	Log-Likelihood:	-76.434
No. Observations:	36	AIC:	156.9
Df Residuals:	34	BIC:	160.0
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	14.0195	0.885	15.840	0.000	12.221	15.818
value_fred_A034RC1A027NBEA	0.0015	0.000	11.748	0.000	0.001	0.002

Omnibus:	2.395	Durbin-Watson:	0.177
Prob(Omnibus):	0.302	Jarque-Bera (JB):	1.478
Skew:	0.221	Prob(JB):	0.478
Kurtosis:	2.111	Cond. No.	1.82e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.82e+04. This might indicate that there are strong multicollinearity or other numerical problems.

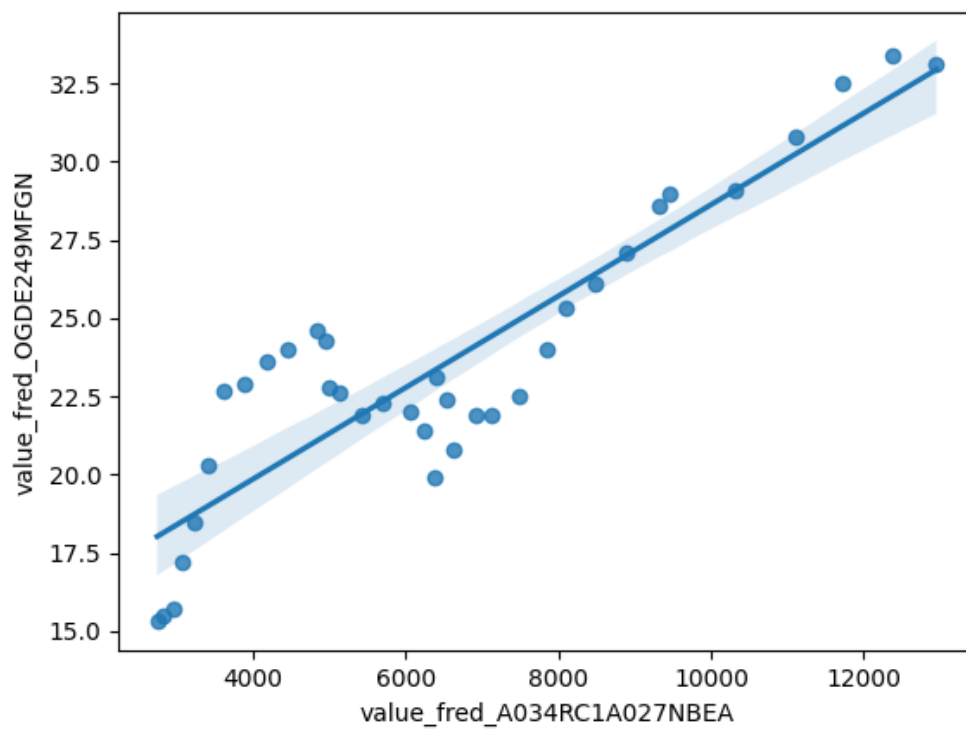


Figure 9: Raw Regression Plot for 2026-09-11

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OGDE249MFGN_detrended	R-squared:	0.401
Model:	OLS	Adj. R-squared:	0.383
Method:	Least Squares	F-statistic:	22.75
Date:	Fri, 11 Sep 2026	Prob (F-statistic):	3.40e-05
Time:	15:19:38	Log-Likelihood:	-74.459
No. Observations:	36	AIC:	152.9
Df Residuals:	34	BIC:	156.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-5.583e-14	0.328	-1.7e-13	1.000	-0.667	0.667
value_fred_A034RC1A027NBEA_detrended	0.0025	0.001	4.770	0.000	0.001	0.004

Omnibus:	0.572	Durbin-Watson:	0.205
Prob(Omnibus):	0.751	Jarque-Bera (JB):	0.649
Skew:	0.067	Prob(JB):	0.723
Kurtosis:	2.356	Cond. No.	638.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

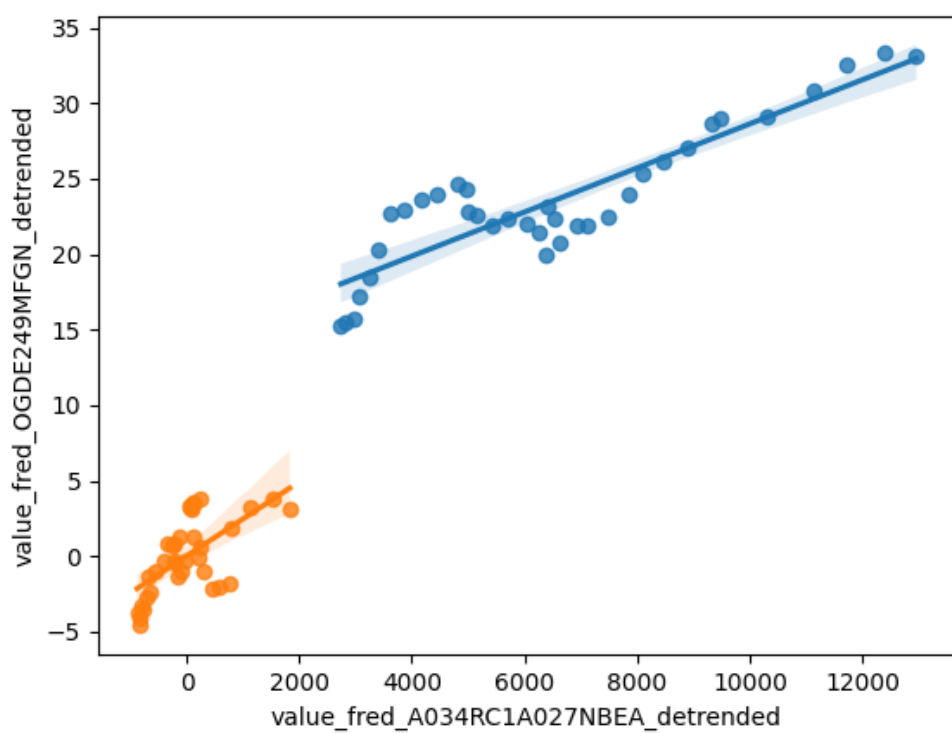


Figure 10: Detrended Regression Plot for 2026-09-11

7 Date: 2026-09-12

Series ID: CUUSA200SAH1

This series is titled Consumer Price Index for All Urban Consumers: Shelter in Midwest - Size Class A and has a frequency of Semiannual. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1984-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

Series ID: APU0400706111

This series is titled Average Price: Chicken, Fresh, Whole (Cost per Pound/453.6 Grams) in the West Census Region - Urban and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1980-01-01 and the observation end date is 2026-08-01. The popularity of this series is 14.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_APU0400706111	R-squared:	0.938			
Model:	OLS	Adj. R-squared:	0.937			
Method:	Least Squares	F-statistic:	1248.			
Date:	Sat, 12 Sep 2026	Prob (F-statistic):	8.79e-52			
Time:	14:28:28	Log-Likelihood:	79.771			
No. Observations:	85	AIC:	-155.5			
Df Residuals:	83	BIC:	-150.7			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.1732	0.034	5.099	0.000	0.106	0.241
value_fred_CUUSA200SAH1	0.0053	0.000	35.326	0.000	0.005	0.006
Omnibus:	3.941	Durbin-Watson:	0.840			
Prob(Omnibus):	0.139	Jarque-Bera (JB):	3.180			
Skew:	0.421	Prob(JB):	0.204			
Kurtosis:	3.436	Cond. No.	742.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

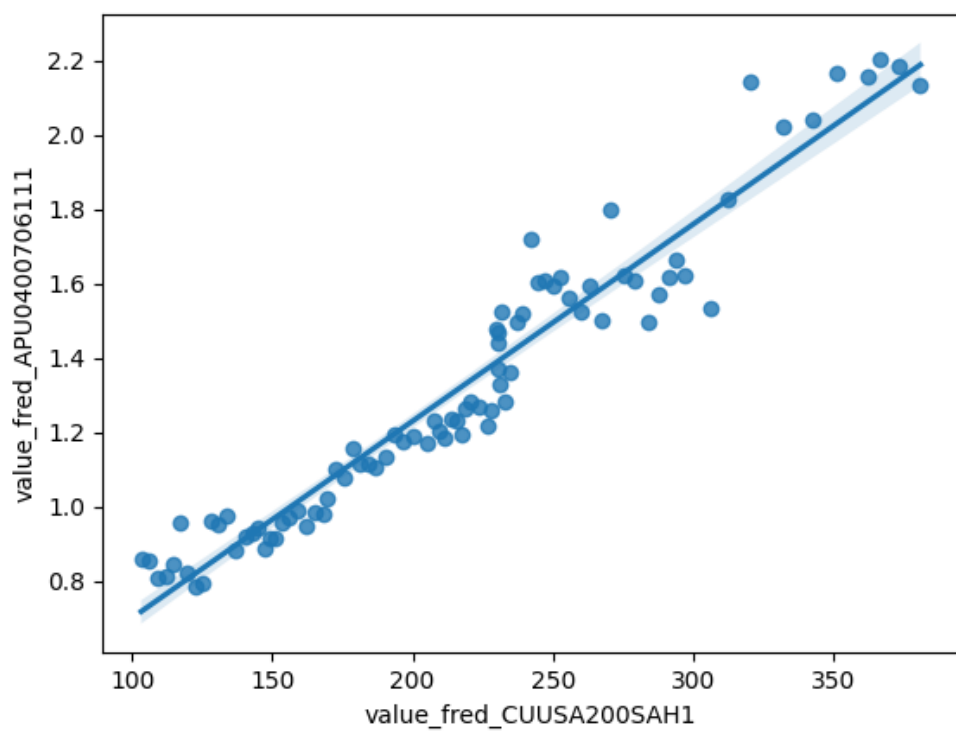


Figure 11: Raw Regression Plot for 2026-09-12

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_APU0400706111_detrended	R-squared:	0.370
Model:	OLS	Adj. R-squared:	0.363
Method:	Least Squares	F-statistic:	48.77
Date:	Sat, 12 Sep 2026	Prob (F-statistic):	6.57e-10
Time:	14:28:28	Log-Likelihood:	79.794
No. Observations:	85	AIC:	-155.6
Df Residuals:	83	BIC:	-150.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.172e-14	0.010	2.09e-12	1.000	-0.021	0.021
value_fred_CUUSA200SAH1_detrended	0.0055	0.001	6.984	0.000	0.004	0.007

Omnibus:	4.280	Durbin-Watson:	0.842
Prob(Omnibus):	0.118	Jarque-Bera (JB):	3.513
Skew:	0.442	Prob(JB):	0.173
Kurtosis:	3.459	Cond. No.	13.3

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

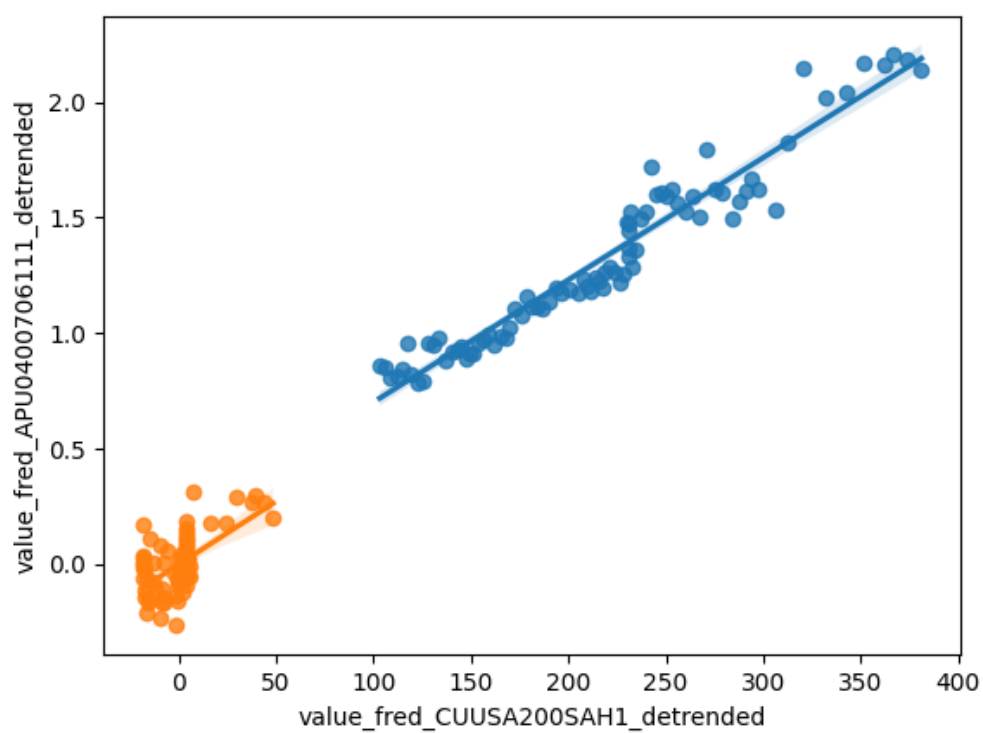


Figure 12: Detrended Regression Plot for 2026-09-12

8 Date: 2026-09-13

Series ID: CIS2010000500000I

This series is titled Employment Cost Index: Total compensation for Private industry workers in Production, transportation, and material moving and has a frequency of Quarterly. The units are Index Dec 2005=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2002-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: ECIGVTCOM

This series is titled Employment Cost Index: Compensation: State and Local Government: All Workers and has a frequency of Quarterly. The units are Index Dec 2005=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2001-01-01 and the observation end date is 2026-04-01. The popularity of this series is 5.

8.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ECIGVTCOM	R-squared:	0.989
Model:	OLS	Adj. R-squared:	0.989
Method:	Least Squares	F-statistic:	8834.
Date:	Sun, 13 Sep 2026	Prob (F-statistic):	2.63e-96
Time:	15:09:51	Log-Likelihood:	-230.11
No. Observations:	98	AIC:	464.2
Df Residuals:	96	BIC:	469.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	7.0722	1.299	5.444	0.000	4.494	9.651
value_fred_CIS2010000500000I	0.9540	0.010	93.989	0.000	0.934	0.974

Omnibus:	47.674	Durbin-Watson:	0.025
Prob(Omnibus):	0.000	Jarque-Bera (JB):	11.046
Skew:	-0.530	Prob(JB):	0.00399
Kurtosis:	1.742	Cond. No.	643.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

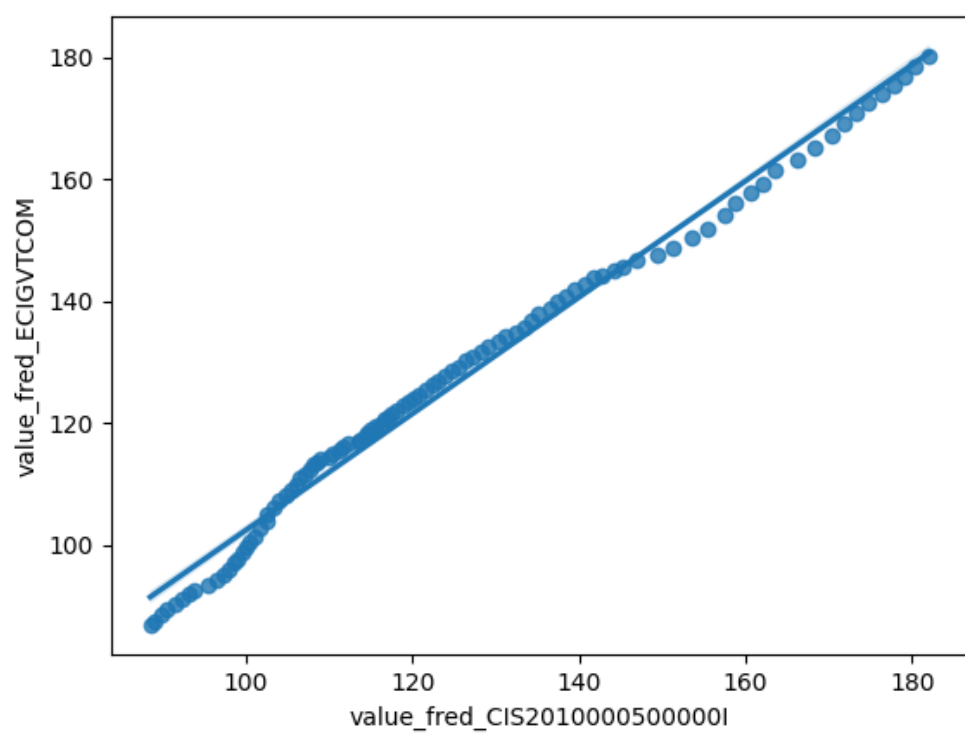


Figure 13: Raw Regression Plot for 2026-09-13

8.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ECIGVTCOM_detrended	R-squared:	0.834
Model:	OLS	Adj. R-squared:	0.832
Method:	Least Squares	F-statistic:	481.9
Date:	Sun, 13 Sep 2026	Prob (F-statistic):	3.38e-39
Time:	15:09:51	Log-Likelihood:	-190.53
No. Observations:	98	AIC:	385.1
Df Residuals:	96	BIC:	390.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.162e-13	0.173	2.41e-12	1.000	-0.343	0.343
value_fred_CIS2010000500000I_detrended	0.6428	0.029	21.952	0.000	0.585	0.701

Omnibus:	4.629	Durbin-Watson:	0.041
Prob(Omnibus):	0.099	Jarque-Bera (JB):	2.345
Skew:	0.034	Prob(JB):	0.310
Kurtosis:	2.245	Cond. No.	5.89

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

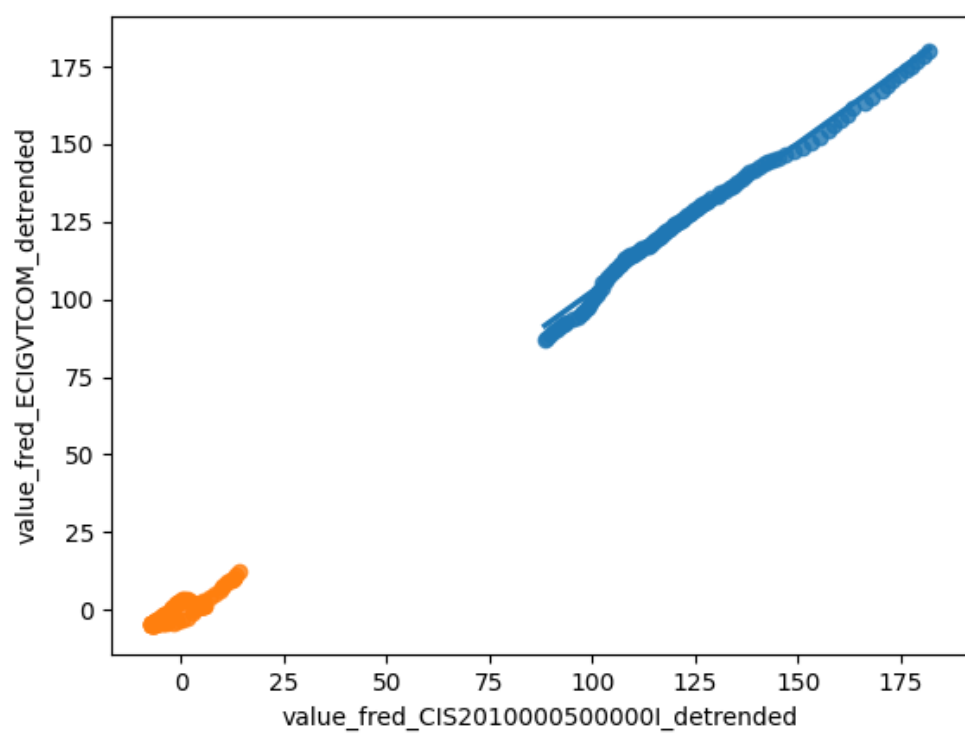


Figure 14: Detrended Regression Plot for 2026-09-13